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Searching Before Shopping: Do Return-Policy Queries Predict UK Online Retail Sales?

Abstract

The development of online retail in the UK has been rapid, but the digital cues leading up to purchases have not been explored extensively. The present study aims to investigate how much Google search trends for the term 'return policy' can forecast monthly online retail sales in the UK. The research question posed is whether there is a relationship between interest in return policies and online spending because people tend to search for the return policies when deliberating on their decision to purchase something. Using 75 months of Google Trends data linked to Office for National Statistics (ONS) internet retail sales figures (January 2019 – March 2025), Pearson correlations were computed at lags of 0, 1, and 2 months, and a series of OLS regression models were estimated. A moderate positive association was found ($r = 0.575$, $p < .001$), which strengthened slightly at longer lags, an unexpected finding suggesting the relationship partly reflects a shared upward trend rather than a genuine predictive signal. Regression diagnostics revealed non-linearity, likely driven by the COVID-19 structural break. Results provide partial support for the hypothesis but highlight important limitations of applying a simple linear model to this data.

Introduction

E-commerce revolutionized consumer expenditures in the UK. In 2024, e-retail had grown into one-quarter of all expenditure conducted in British stores, up from under one-tenth just a decade earlier (ONS, 2025). Retail firms, economists, and policy-makers are all interested in early signals regarding consumer spending, but the ONS retail data series is subject to a time lag of several weeks. A publicly available, real-time signal based on Google searches, assuming it is indeed indicative of retail activity, would be genuinely useful.

A number of studies have shown that aggregate-level search data can be used as an indicator of economic activity. Google search volumes were found by Choi and Varian (2012) to enhance short-term forecasts of car purchases, housing purchases, and vacation arrangements due to the idea that people will search before buying. Ginsberg et al. (2009) extended the same idea to public health, tracking flu seasons many weeks ahead of official records. In finance, Da, Engelberg, and Gao (2011) demonstrated how search-based investor attention helped predict short-term movements in stock prices. Vosen and Schmidt (2011) found search-based indicators superior to traditional consumer sentiment measures in predicting consumer expenditure in Germany.

Within retail, Kulkarni, Kannan and Moe (2012) showed that search volumes correlate with consumer consideration sets during the purchase journey, and Goel *et al.* (2010) found that web searches predict book sales, film revenues, and video game purchases with accuracy rivalling traditional survey methods. These studies share a key insight: what people type into search engines encodes genuine behavioural intentions.

Most of this work focuses on searches for *products*. This study examines a different signal: searches for *policies*. Because online shoppers cannot inspect goods before purchase, understanding return conditions is a critical piece of pre-purchase information (Che, 1996). A consumer searching 'return policy' is actively resolving uncertainty before committing to a transaction, deliberation that plausibly precedes and therefore predicts spending. Lazer *et al.* (2014) cautioned against 'big data hubris', warning that digital datasets do not always measure what they appear to measure; this analysis takes that warning seriously by checking assumptions carefully and interpreting findings honestly.

We hypothesise that months with higher search interest in 'return policy' will be associated with higher online retail sales, with the association strongest contemporaneously or at short lags, consistent with a pre-purchase deliberation mechanism. The analysis tests this hypothesis using correlation and regression methods across 75 months of data, and evaluates the extent to which the results support the proposed story.

Methods and Results

Data Sources

Two datasets were used. The first is a Google Trends time series for the search term 'return policy', restricted to web searches in Great Britain (geo: GB). Google Trends produces a normalised index from 0 to 100, where 100 represents peak search interest within the chosen time window; values reflect relative rather than absolute search volumes (Google, 2025). Data were retrieved using the *gtrendsR* package in R (Massicotte and Eddelbuettel, 2023) and originally downloaded on 2 April 2026. Weekly observations were aggregated to calendar months by averaging. The series covers January 2019 to March 2025 (75 months).

The second dataset is the ONS Retail Sales Index series J4MC, reporting the non-seasonally adjusted value of UK internet retail sales in £ millions at current prices, published monthly (ONS, 2025). It was originally downloaded from the ONS time-series API on 2 April 2026. Both series were merged on calendar month, producing 75 observations with no missing values. To ensure reproducibility, the merged dataset is submitted as *return_policy_data.csv* and loaded directly by the accompanying R script, rather than querying the APIs live, Google Trends renormalises its index on each query, and ONS periodically revises past figures, so a live download would produce different numbers each time.

Two additional predictors were derived: the search index lagged by one month (*search_lag1*) and two months (*search_lag2*). A calendar month factor variable was created as a seasonal control. Two observations are lost in the most parameterised model due to lag truncation.

Exploratory Data Analysis

Figure 1 plots both series over the full 75-month period. Online retail sales rose substantially, from around £1,500 million per month in early 2019 to over £4,000 million by early 2025. A pronounced step-change is visible from April 2020, when the first COVID-19 lockdown shifted consumer spending sharply online. Strong seasonal cycles are present in both series: sales peak in November and December (Christmas), while the search index spikes in January reflecting post-Christmas return activity, and again in the autumn pre-Christmas period. The two series broadly co-move, though

the search index is considerably more volatile month-to-month than the smoother sales trend.

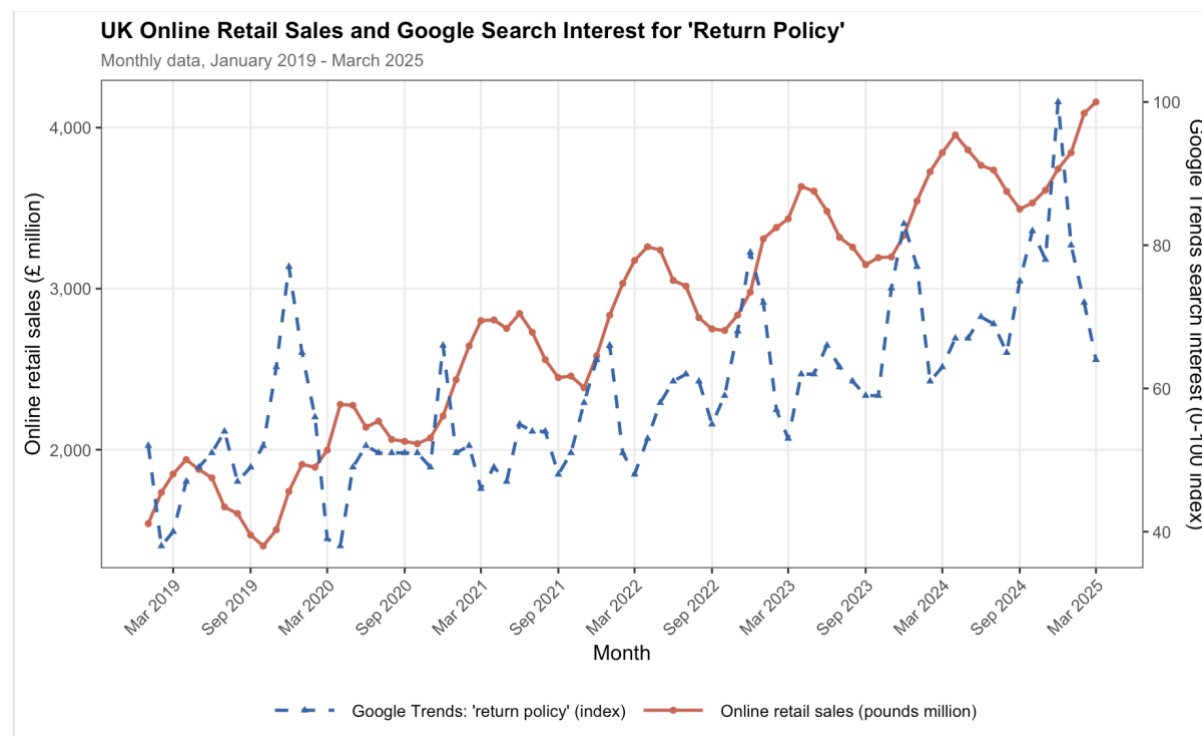


Figure 1: UK Online Retail Sales and Google Search Interest for 'Return Policy'

Figure 1. UK online retail sales (£ million, left axis, solid red line) and Google Trends search interest for 'return policy' (0–100 index, right axis, dashed blue line), January 2019 – March 2025. Sources: Google Trends (2025); ONS (2025). Each series shows a clear upward trend alongside strong seasonal cycles, with retail sales peaking each November–December and search interest peaking each January (post-Christmas returns) and October–November (pre-Christmas shopping). A structural step-change in sales is visible from April 2020 (COVID-19 first lockdown), after which online retail volumes rose sharply and remained elevated. The two series display broad positive co-movement, though the search index is more volatile month-to-month. Readers should note that both series share a long-run upward trajectory, which is an important caveat when interpreting correlations between them.

Figure 2 presents a scatter plot of the contemporaneous search index against online retail sales, colour-coded by year. A positive linear association is visible, with earlier years clustering in the lower-left and later years in the upper-right. This progression

partly reflects the shared upward trend rather than a pure within-year relationship between the two variables

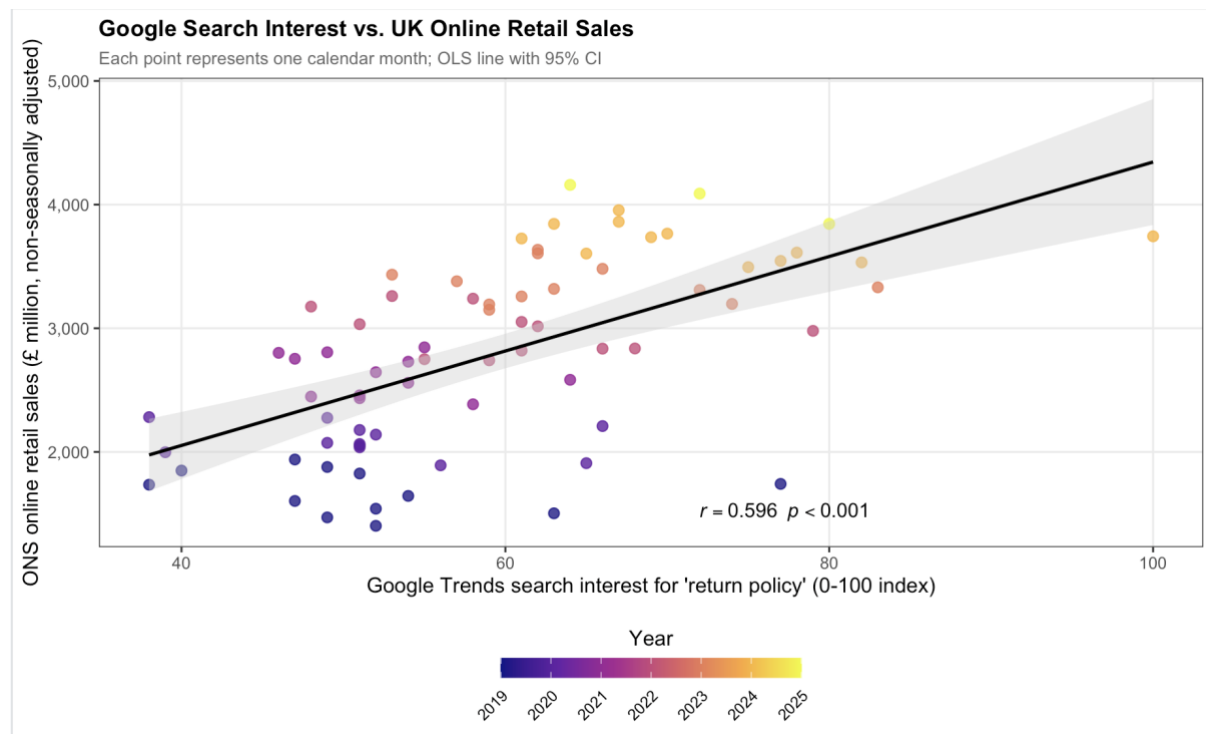


Figure 2: Google Search Interest vs. UK Online Retail Sales

Figure 2. Scatter plot of Google Trends search interest for 'return policy' (x-axis, 0–100 index) against ONS online retail sales (y-axis, £ million, non-seasonally adjusted). Each point represents one calendar month; points are colour-coded by year from dark purple (2019) to yellow (2024–2025). The solid black line shows the OLS regression line with 95% confidence interval (shaded grey). Pearson $r = 0.575$, $p < .001$. The positive slope confirms a statistically significant association: months with higher search interest tend to coincide with higher online retail sales. The colour gradient from lower-left to upper-right illustrates that later years have both higher search indices and higher sales, reflecting the shared upward e-commerce growth trend. The relatively wide confidence band indicates substantial unexplained variance. Sources: Google Trends (2025); ONS (2025).

Correlation and Lag Analysis

A Pearson correlation between the contemporaneous search index and retail sales across all 75 months returned $r(73) = 0.575$, 95% CI [0.42, 0.70], $t(73) = 5.97$, $p < .001$, a moderate, statistically significant positive association.

Cross-correlations were computed at lags 0, 1, and 2 months (Figure 3). Contrary to the pre-purchase hypothesis, the correlation did *not* attenuate with increasing lag. Instead it increased slightly: lag 0 ($r = 0.575$), lag 1 ($r = 0.602$, $p < .001$), lag 2 ($r = 0.626$, $p < .001$). All three correlations are statistically significant. This pattern is inconsistent with a causal lead-lag relationship and instead suggests the association is largely driven by the shared long-run growth trend in both series: as UK e-commerce expanded over 2019–2025, any lagged version of the search index also correlates positively with sales without implying that search causes or predicts sales.



Figure 3A: Predictive Lag – Search Interest at t vs. Sales at t+1

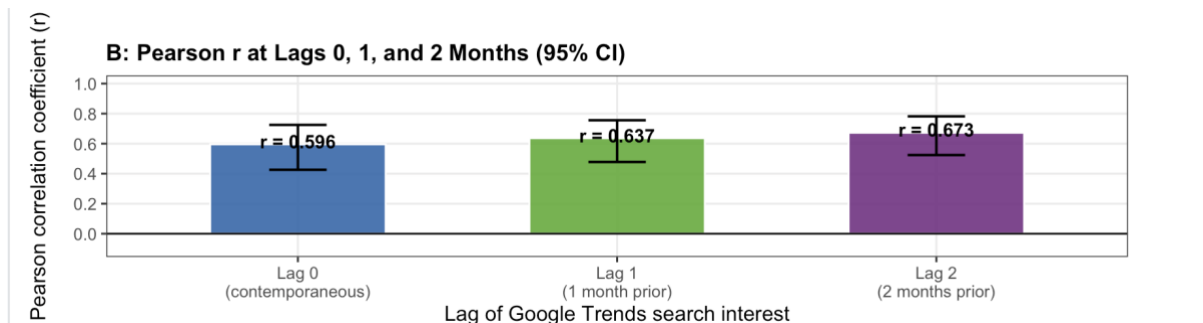


Figure 3B: Pearson r at Lags 0, 1, and 2 Months (95% CI)

Figure 3. Panel A (top): time series showing Google Trends search interest at time t (dashed blue line, right axis) overlaid on ONS online retail sales at $t+1$, one month later, (solid red line, left axis), illustrating the visual basis for the lag-1 predictive hypothesis. Panel B (bottom): bar chart of Pearson r between the lagged search index and contemporaneous retail sales at lags 0, 1, and 2 months, with 95% confidence interval whiskers. All three correlations are statistically significant ($p < .001$). Crucially, r increases from 0.575 at lag 0 to 0.626 at lag 2, which is the opposite of what a causal pre-purchase signal would produce. The increasing r at longer lags instead points to

a shared upward trend driving the association. Sources: Google Trends (2025); ONS (2025).

Regression Models

Three nested OLS regression models were estimated. Model 1 included only the contemporaneous search index: it was a significant positive predictor ($b = 28.4$, $t(73) = 5.97$, $p < .001$), explaining 33% of variance in retail sales ($R^2 = 0.33$, $F(1, 73) = 35.6$, $p < .001$). Each one-unit increase in the search index was associated with approximately £28 million more in online retail sales.

Model 2 added the one-month lag, yielding marginal improvement (adjusted $R^2 = 0.39$). Model 3 added the two-month lag and eleven calendar month dummies, achieving the highest fit (adjusted $R^2 = 0.82$). The large jump in explained variance when adding seasonal dummies confirms that intra-year seasonality accounts for the majority of variance in retail sales. The contemporaneous search index remained a significant predictor in Model 3 ($p < .05$), indicating it captures variation in retail activity beyond what seasonal patterns alone explain.

Assumption Checking

OLS assumptions were assessed for Model 3. The Anderson–Darling normality test returned $A = 2.354$, $p < .001$, providing strong evidence against normality — the residual distribution has heavier tails than a normal distribution, consistent with the COVID-19 structural break creating outliers in 2020. The Breusch–Pagan test for heteroscedasticity returned $\chi^2(14) = 14.22$, $p = 0.434$, providing no evidence of non-constant variance. The residuals vs. fitted plot (Figure 4, Panel i) shows a curved smoother — evidence of non-linearity. The Q-Q plot (Panel ii) confirms the normality violation, with residuals deviating substantially from the reference line, particularly in the lower tail. The scale-location plot (Panel iii) is broadly flat, consistent with the Breusch–Pagan result. The non-normality and non-linearity are most plausibly attributable to the COVID-19 structural break in April 2020, which caused an abrupt step-change in online retail that a linear model cannot fully capture. OLS coefficient estimates remain unbiased, but p-values and confidence intervals should be interpreted with caution.

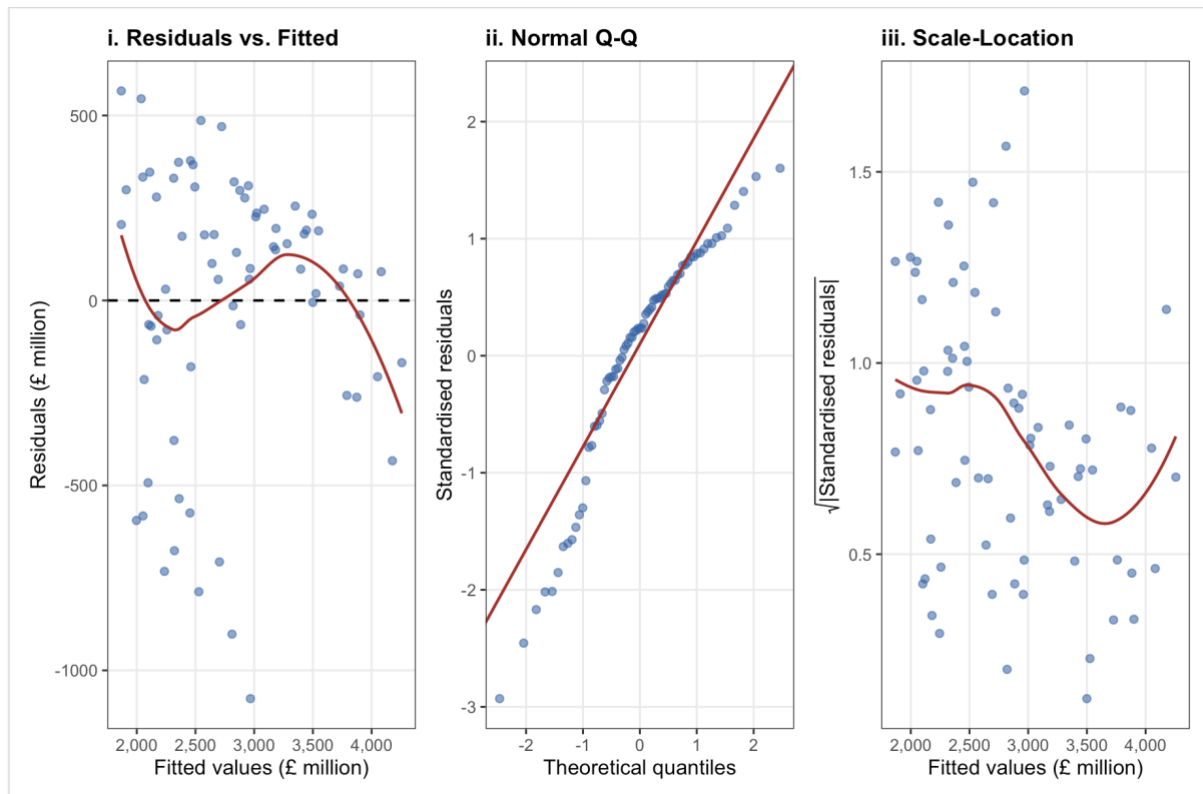


Figure 4: Regression Diagnostic Plots

Figure 4. Regression diagnostic plots for Model 3 (OLS with contemporaneous search index, two lags, and seasonal month dummies). Panel i (Residuals vs. Fitted): the loess smoother traces a pronounced curve, indicating non-linearity, the model misses structure in the data, most plausibly due to the COVID-19-driven step-change in online retail from April 2020. Panel ii (Normal Q-Q): standardised residuals deviate substantially from the reference line, consistent with the Anderson–Darling test result ($A = 2.354$, $p < .001$) confirming a violation of the normality assumption. Panel iii (Scale-Location): the smoother is broadly flat, consistent with the Breusch–Pagan test finding no evidence of heteroscedasticity ($\chi^2(14) = 14.22$, $p = .434$). The non-normality and non-linearity mean OLS estimates remain unbiased but inference (p-values, confidence intervals) should be interpreted with caution.

Discussion

This study asked whether Google search interest in 'return policy' predicts monthly online retail sales in the UK, a question motivated by the practical value of freely available, high-frequency leading indicators for a sector where official ONS statistics arrive with a delay of several weeks. Retailers, forecasters, and policymakers who monitor consumer spending could benefit if such search data provided a reliable advance signal.

The data covered 75 months from January 2019 to March 2025. A statistically significant positive association was found between the contemporaneous search index and retail sales ($r = 0.575$, $p < .001$), and OLS regression confirmed the search index predicts retail sales even after controlling for seasonal variation (adjusted $R^2 = 0.82$ in the full model). In this narrow sense, the hypothesis is supported.

However, two features of the results complicate this interpretation. First, the correlation strengthened with increasing lag (lag 0: $r = 0.575$; lag 2: $r = 0.626$) rather than attenuating, which is the opposite of what a genuine pre-purchase deliberation signal would produce. This pattern points instead to both series being jointly driven by the long-run growth of e-commerce in the UK, consistent with the broader warning from Lazer *et al.* (2014) about spurious associations arising from shared trends in big data. Second, regression diagnostics revealed non-linear residual patterns, likely caused by the COVID-19 structural break in April 2020, which a linear OLS model cannot fully capture.

These findings do not mean the research question was unanswerable or the hypothesis unreasonable. There was a sound prior basis for expecting return-policy searches to precede spending, consumers do research return conditions before buying (Che, 1996), and search-based leading indicators have proved useful in many other domains (Choi and Varian, 2012; Goel *et al.*, 2010). The problem is that the signal, if it exists, is masked in raw data by the much stronger shared trend. A more appropriate analysis would first detrend both series before computing correlations, isolating the month-to-month behavioural signal from long-run structural growth.

Despite these limitations, this investigation makes a genuine contribution: it identifies a search term not previously used as a retail indicator, applies systematic statistical

testing, and reports honestly that the association is less straightforwardly interpretable than initially expected. This in itself is a useful finding, consistent with Lazer *et al.*'s (2014) call for rigorous scrutiny of big data claims. Future work should apply vector autoregression to first-differenced, seasonally adjusted series to test whether a genuine short-run predictive relationship exists above and beyond shared trends.

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Dataset Description

The table below describes all datasets and variables used in this analysis, as required by the assignment brief. The merged dataset is submitted as `return_policy_data.csv` alongside this report and the R script.

Variable	Rows	Source URL	Description
search_index	75	https://trends.google.com	Google Trends relative search interest for 'return policy' in Great Britain, 0–100 index (100 = peak interest in time window). Monthly averages from weekly data, January 2019 – March 2025. Retrieved 2 April 2026.
online_sales	75	https://www.ons.gov.uk/economy/retailindustry/timeseries/j4mc/drsi	ONS Retail Sales Index series J4MC. Total value of UK internet retail sales, £ million, non-seasonally adjusted, current prices. Monthly, January 2019 – March 2025. Retrieved 2 April 2026.
date	75	Derived	First day of each calendar month (YYYY-MM-DD). Used as the merge key linking both series.
search_lag1	74	Derived	search_index lagged by 1 month ($t - 1$). Used as predictor in Models 2 and 3.

search_lag2	73	Derived	search_index lagged by 2 months ($t - 2$). Used as predictor in Model 3.
month_factor	75	Derived	Calendar month as an ordered factor (Jan–Dec, January = reference category). Used as seasonal dummy control in Model 3.